PiyaNota: A Mobile Application for Optical Music Recognition and Piano Audio Rendering of Filipino SATB Choral Sheet Music

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Abstract. Choral singing is prevalent in Filipino culture, practiced in worship services, schools, and community choirs, yet many choristers have limited music-reading skills and learn parts from conductors. Existing tools fall short: Optical Music Recognition (OMR) models degrade on photos, commercial apps assume musical literacy, SkiMusic, a Filipino-developed system, applies fixed rules to piano scores only, and no annotated Filipino SATB dataset exists. We present PiyaNota, an Android app converting photos of open-score SATB (Soprano, Alto, Tenor, Bass) scores into per-voice piano audio. YOLOv8 isolates staves; a convolutional recurrent network fine-tuned on multi-font renderings, with a clef and key-signature classifier, transcribes them. In a 12-page pilot, all 100 staves were detected; camera pitch accuracy rose from under 5% to 83.3–83.4% (88.7–89.1% digital), though camera symbol error rate remains 33.8–34.8%. PiyaNota contributes a dataset, a deep learning pipeline, and a step toward musical literacy for singers who learn by ear.

Keywords: Optical Music Recognition, Computer Vision, Deep Learning, Mobile Application, Filipino Choral Music, Musical Literacy

Session: Computation for Community-based Applications

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Supporting Material: PiyaNota Work-in-Progress Status

1. System Overview and Development Status

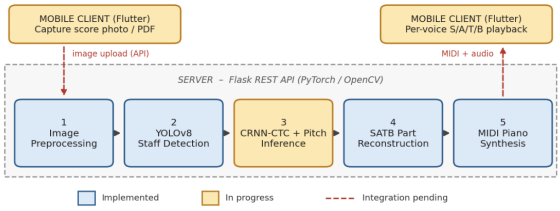


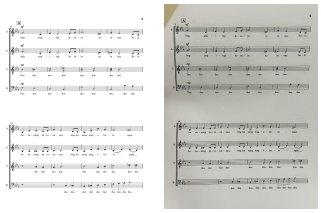
Fig. 1. PiyaNota pipeline: mobile capture, five-stage server-side processing, and per-voice audio playback.

2. Preliminary Results: Camera-Captured vs. Digital Input

Table 1. Final system (fine-tuned CRNN-CTC + clef/key classifier) on a 12-page pilot (100 staves, 1,008 notes per condition). Ranges span two training seeds. Staff detection = share of staves found and assigned the correct voice; pitch accuracy ignores note duration; lower SER is better.

Input Condition	Staff Detection (recall)	Symbol Error Rate (SER)	Pitch Inference Acc.
Digital (PDF render)	100.0%	25.0–25.5%	88.7–89.1%
Camera-captured photo	100.0%	33.8–34.8%	83.3–83.4%
Degradation (Δ)	0.0 pp	+8.8 to +9.2 pp	−5.4 to −5.7 pp

The initial pipeline (CRNN-CTC trained only on single-font synthetic data) detected 84.0% of staves on digital and 56.0% on camera input, with pitch accuracy below 5% in both because it read no clef correctly. Pitch accuracy does not penalise extra notes: about one in three predicted notes is spurious or wrong-pitched (note precision 66–69%).



(a) (b)

Fig. 2. The same pilot page (*Gaano Ko Ikaw Kamahal*, p. 2) as digital render (a) and smartphone capture (b), showing lighting, skew, and noise artifacts.

3. Remaining Work

Remaining work involves (1) reducing spurious notes and rhythm errors, which drive the remaining symbol error, and raising camera pitch accuracy toward the 85% target; (2) evaluating on a larger held-out set of Filipino SATB scores; (3) completing the Flutter mobile application and connecting it to the Flask backend; and (4) a usability evaluation with Filipino choir members using the System Usability Scale.